Reorganized Chapter 3: Geometric Class Field Theory

A standalone review copy for comparison with the current chapter
(newly written mathematics in blue)

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Chapter 3

Geometric Class Field Theory

The goal of this chapter is to prove the reduced form of geometric class field theory stated in the Introduction, which we now recall:

Theorem 1 (Geometric Class Field Theory, reduced form; restated from Theorem 2)**.** *Let Λ be a finite ring of cardinality invertible in k, and let $\mathcal{F}$ be an étale sheaf of Λ-modules, locally free of rank 1 on $U_{\acute{e}t}$, with ramification bounded by $\mathfrak{m}$. Then, for sufficiently large integer d, there exists a unique (up to isomorphism) étale sheaf of Λ-modules $\mathcal{G}_d$ on $\mathrm{Pic}^d_{C,\mathfrak{m}}$, locally free of rank 1, such that the pullback of $\mathcal{G}_d$ by Φ_d is isomorphic to $\mathcal{F}^{[d]}$.*

Throughout the chapter k is a perfect field, C is a projective smooth geometrically connected curve over k of genus g, $\mathfrak{m}$ is a modulus on C with complement $U = C \setminus \mathfrak{m}$, and $\Phi_d : U^{(d)} \to \mathrm{Pic}^d_{C,\mathfrak{m}}$ is the degree d Abel-Jacobi morphism. The strategy, going back to Deligne, is to descend the symmetric power $\mathcal{F}^{[d]}$ along Φ_d.

The unramified case. When $\mathfrak{m} = 0$ we have $U^{(d)} = C^{(d)}$, and for d large enough Φ_d is proper, smooth and surjective, with geometrically connected fibers isomorphic to projective spaces. Projective spaces are simply connected, so the homotopy exact sequence for the étale fundamental group (recalled in Section 3.1) shows that Φ_d induces an isomorphism $\pi_1^{et}(C^{(d)}) \cong \pi_1^{et}(\mathrm{Pic}^d_C)$; every local system on $C^{(d)}$ then descends, uniquely, to Pic^d_C.

The ramified case. When $\mathfrak{m} > 0$ the fibers of Φ_d are affine spaces and Φ_d is no longer proper, so the homotopy argument does not apply directly. Following Takeuchi [Tak19], we repair this by compactifying the fibers: inside the blowup $X_{\mathfrak{m},d}$ of $C^{(d)}$ along the locus Z_0 of divisors containing $\mathfrak{m}$, Takeuchi constructs an open subscheme $\tilde{C}^{(d)}_{\mathfrak{m}} \supseteq U^{(d)}$ to which Φ_d extends as a projective space bundle

$$\tilde{\Phi}_d : \tilde{C}^{(d)}_{\mathfrak{m}} \to \mathrm{Pic}^d_{C,\mathfrak{m}},$$

whose boundary $H = \tilde{C}^{(d)}_{\mathfrak{m}} \setminus U^{(d)}$ is supported on the exceptional divisor of the blowup (Section 3.4). The homotopy argument now applies to $\tilde{\Phi}_d$, and the problem becomes moving $\mathcal{F}^{[d]}$ from $U^{(d)}$ to

$\tilde{C}^{(d)}_{\mathfrak{m}}$, i.e. extending it across the boundary H:

$$\begin{tikzcd}
\mathcal{F}^{[d]} \arrow[d] & \tilde{\mathcal{F}}^{[d]} \arrow[d, dashed] & \\
U^{(d)} \arrow[r, hook] \arrow[dr, "\Phi_d"'] & \tilde{C}^{(d)}_{\mathfrak{m}} \arrow[d, "\tilde{\Phi}_d"] & H = \tilde{C}^{(d)}_{\mathfrak{m}} \setminus U^{(d)} \arrow[l, hook'] \\
& \mathrm{Pic}^d_{C,\mathfrak{m}} &
\end{tikzcd}$$

(The purple arrows emphasize that they are morphisms of sheaves on the étale site; the dashed one is the extension we seek.)

This splits the proof into two tasks:

Task (A): tameness along the boundary. The extension is governed by the ramification of $\mathcal{F}^{[d]}$ at the boundary. The technical main theorem of this chapter asserts that this ramification is as mild as possible:

Theorem 2. *Let Λ be a finite ring of cardinality invertible in k, and let $\mathcal{F}$ be an étale sheaf of Λ-modules, locally free of rank 1 on U, with ramification bounded by $\mathfrak{m}$. Considering $U^{(d)}$ as an open subscheme of the blowup $\tilde{C}^{(d)}_{\mathfrak{m}}$ of $C^{(d)}$, we have that for sufficiently large integer d, $\mathcal{F}^{[d]}$ is tamely ramified on $H = \tilde{C}^{(d)}_{\mathfrak{m}} \setminus U^{(d)} = E_0 \times_{C^{(d)}} \tilde{C}^{(d)}_{\mathfrak{m}}$.*

This is where the work of Chapter 2 enters: there we proved (Theorem 3) that for a *single* point P of the modulus, with $\mathfrak{n} = n_P P \subset \mathfrak{m}$, the symmetric power $\mathcal{F}^{[\deg \mathfrak{n}]}$ is tamely ramified at the generic point of the exceptional divisor of the blowup of $C^{(\deg \mathfrak{n})}$ at $\mathfrak{n}$. In Section 3.5 we assemble these single-point statements into Theorem 2: first combining the points of the modulus with one another, then combining the modulus with the remaining $d - \deg \mathfrak{m}$ "free" points.

Task (B): extension and descent. Tame ramification along H guarantees that $\mathcal{F}^{[d]}$ extends to a locally constant sheaf on all of $\tilde{C}^{(d)}_{\mathfrak{m}}$ (Lemma 15); the extension then descends along $\tilde{\Phi}_d$ by the isomorphism of fundamental groups, and restricting back to $U^{(d)}$ proves Theorem 1. This is carried out in Section 3.6.

Organization of the chapter.

Section 3.1 recalls the homotopy exact sequence for étale fundamental groups, and the notion of tame fundamental groups, which is the reason tameness is the right condition in Task (A).

Section 3.2 recalls the generalized Picard scheme $\mathrm{Pic}^d_{C,\mathfrak{m}}$, the target of the descent.

Section 3.3 recalls the Abel-Jacobi morphism Φ_d and the structure of its fibers.

Section 3.4 recalls Takeuchi's compactification $\tilde{C}^{(d)}_{\mathfrak{m}}$ and the extended morphism $\tilde{\Phi}_d$.

Section 3.5 proves Theorem 2 (Task (A)) in four steps: an auxiliary analysis of ramification on products of blowups; étaleness of the relevant addition maps between symmetric powers; spreading tameness from the blowup at the modulus to the full exceptional divisor E_0; and the assembly of the proof.

Section 3.6 combines everything into the proof of Theorem 1 (Task (B)).

3.1 The Homotopy Exact Sequence and Tame Fundamental Groups

We recall the definition and basic properties of the étale fundamental group, following the Stacks Project [Stacks, Tag 0BQ6].

Proposition 3 ([Stacks, Tag 0C0J])**.** *Let $f : X \to S$ be a flat proper morphism of finite presentation whose geometric fibres are connected and reduced. Assume S is connected and let $\overline{s}$ be a geometric point of S. Then there is an exact sequence*

$$\pi_1(X_{\overline{s}}) \to \pi_1(X) \to \pi_1(S) \to 1$$

of fundamental groups.

Corollary 4. *Let $f : X \to S$ be a proper smooth morphism of finite presentation whose geometric fibres are connected. Assume S is connected and let $\overline{s}$ be a geometric point of S. Then there is an exact sequence*

$$\pi_1(X_{\overline{s}}) \to \pi_1(X) \to \pi_1(S) \to 1$$

of fundamental groups.

Proof. A smooth morphism is flat, and its geometric fibers are regular, hence reduced; so f satisfies the hypotheses of Proposition 3. $\square$

We will apply Corollary 4 to fibrations in projective spaces. Since $\mathbb{P}^n_{k^{sep}}$ is simply connected, in that case the sequence collapses to an isomorphism $\pi_1(X) \xrightarrow{\sim} \pi_1(S)$: finite étale covers of X and of S (equivalently, by Proposition 9, torsors and rank-one local systems) correspond uniquely to one another by pullback and descent.

We also recall why *tameness* is the relevant condition in this chapter. Let X be a normal integral scheme and let $U \subset X$ be a dense open subscheme. Recall (Definition 38) that a G-torsor on U_{et} is said to be tamely ramified at the generic point ξ of a prime divisor $D \subset X$ with $D \cap U = \emptyset$ if the corresponding extensions of the discrete valuation ring $\mathcal{O}_{X,\xi}$ are tamely ramified, i.e. have ramification index invertible in the residue field and separable residue field extensions. Covers of U that are tamely ramified in codimension 1 with respect to X form a Galois subcategory of the finite étale covers of U, and the corresponding quotient of $\pi_1^{et}(U)$ is the *tame fundamental group* of the pair $U \subset X$ (see [Stacks, Tag 0EYD] and the surrounding sections for this formalism). The relevance for us: a local system on U extends to a locally constant sheaf on X if and only if its monodromy factors through $\pi_1^{et}(X)$, and tame ramification along the boundary is precisely the local condition that makes such an extension possible in our situation — this is the content of Lemma 15 below, and the reason Task (A) asks for tameness and nothing finer.

3.2 The Generalized Picard Scheme

We recall the notion of generalized Jacobian varieties and study their fundamental properties. The material presented here is primarily adapted from [Gui19] and [Tak19]. For further background on the general theory of abelian varieties and Jacobians, the reader may also consult [Mil08]. Let S be a scheme and let C be a projective smooth S-scheme whose geometric fibers are connected and of dimension 1. Let $\mathfrak{m}$ be a modulus on C, defined as an effective Cartier divisor of C/S (i.e., a closed subscheme of C which is finite flat of finite presentation over S). We denote the projection $C \times_S T \to T$ by pr for any S-scheme T.

The Functor of Points

Let d be an integer. For an S-scheme T, we consider the set of data $(\mathcal{L}, \psi)$ where:

- $\mathcal{L}$ is an invertible sheaf of degree d on C_T.
- $\psi : \mathcal{O}_{\mathfrak{m}_T} \xrightarrow{\sim} \mathcal{L}|_{\mathfrak{m}_T}$ is a trivialization of $\mathcal{L}$ along the modulus.

Two such pairs $(\mathcal{L}, \psi)$ and $(\mathcal{L}', \psi')$ are said to be isomorphic if there exists an isomorphism of invertible sheaves $f : \mathcal{L} \to \mathcal{L}'$ such that the following diagram commutes:

$$\begin{tikzcd} & \mathcal{O}_{\mathfrak{m}_T} \arrow[dl, "\psi'"'] \arrow[dr, "\psi"] & \\ \mathcal{L}'|_{\mathfrak{m}_T} \arrow[rr, "f|_{\mathfrak{m}_T}"] & & \mathcal{L}|_{\mathfrak{m}_T} \end{tikzcd}$$

We define the presheaf $\mathrm{Pic}^{d,\mathrm{pre}}_{C,\mathfrak{m}}$ on Sch/S by assigning to T the set of isomorphism classes of such pairs. Let $\mathrm{Pic}^d_{C,\mathfrak{m}}$ denote the étale sheafification of this presheaf.

Representability and Structure

The fundamental properties of this functor are as follows:

1. $\mathrm{Pic}^d_{C,\mathfrak{m}}$ is represented by an S-scheme. (Note: If $\mathfrak{m}$ is faithfully flat over S, the presheaf is already an étale sheaf).
2. $\mathrm{Pic}^0_{C,\mathfrak{m}}$ is a smooth commutative group S-scheme with geometrically connected fibers, referred to as the *generalized Jacobian variety* of C with modulus $\mathfrak{m}$.
3. For any d, $\mathrm{Pic}^d_{C,\mathfrak{m}}$ is a $\mathrm{Pic}^0_{C,\mathfrak{m}}$-torsor.

In the case where $\mathfrak{m} = 0$, we recover the standard Jacobian variety, denoted simply as Pic^d_C.

Remark (Sketch of representability). We indicate why these properties hold, following [Gui19, Section 4].

(i) *Rigidified pairs have no automorphisms.* Assume $\mathfrak{m}$ is faithfully flat over S. An automorphism of a pair $(\mathcal{L}, \psi)$ over T is a global unit $u \in \Gamma(C_T, \mathcal{O}_{C_T})^\times = \Gamma(T, \mathcal{O}_T)^\times$ (the equality because $\mathrm{pr}_*\mathcal{O}_{C_T} = \mathcal{O}_T$, as $C_T \to T$ is proper with geometrically connected and reduced fibers) whose restriction to $\mathfrak{m}_T$ is the identity. Since $\mathfrak{m}_T \to T$ is faithfully flat, $u = 1$. Having no nontrivial automorphisms, isomorphism classes of pairs glue in the étale topology, so the presheaf is already a sheaf.

(ii) *An exact sequence.* Forgetting ψ and comparing two trivializations of the same $\mathcal{L}$ yields an exact sequence of étale sheaves of abelian groups

$$1 \longrightarrow \left(\mathrm{pr}_*\mathcal{O}_{\mathfrak{m}}^\times\right)/\mathcal{O}_S^\times \longrightarrow \mathrm{Pic}^0_{C,\mathfrak{m}} \longrightarrow \mathrm{Pic}^0_C \longrightarrow 1,$$

where the kernel $T_{\mathfrak{m}} := \big(\mathrm{pr}_*\mathcal{O}_{\mathfrak{m}}^\times\big)/\mathcal{O}_S^\times$ is the quotient of the Weil restriction of $\mathbb{G}_m$ along the finite flat morphism $\mathfrak{m} \to S$ by the diagonal $\mathbb{G}_m$: a smooth affine group S-scheme with geometrically connected fibers. (Over $S = \operatorname{Spec} k$ with $\mathfrak{m} = \sum n_i P_i$, its fibers are products of extensions of the tori $\mathrm{Res}_{\kappa(P_i)/k}\mathbb{G}_m$ by unipotent groups recording the higher congruences along $n_i P_i$, of total dimension $\deg \mathfrak{m} - 1$.)

(iii) *Conclusion.* Pic^0_C is representable by an abelian S-scheme (Grothendieck; see [Mil08]). The sequence in (ii) exhibits $\mathrm{Pic}^0_{C,\mathfrak{m}}$ as a $T_{\mathfrak{m}}$-torsor over Pic^0_C; since $T_{\mathfrak{m}}$ is affine, such a torsor is representable by descent, and smoothness and geometric connectedness of the fibers follow from the corresponding properties of $T_{\mathfrak{m}}$ and Pic^0_C. This proves (1) and (2) for $d = 0$; (3) follows since tensoring by a degree d pair identifies the degree 0 and degree d functors, and representability of $\mathrm{Pic}^d_{C,\mathfrak{m}}$ follows from (3) by the same descent argument.

Relation to the Standard Jacobian

We now examine the behavior of the generalized Picard scheme under the variation of the modulus. By viewing the structure along the modulus as an additional rigidification, we obtain natural transition maps corresponding to the inclusion of moduli.

Let $\mathfrak{m}_1$ and $\mathfrak{m}_2$ be moduli such that $\mathfrak{m}_1 \subset \mathfrak{m}_2$. There exists a natural map

$$\mathrm{Pic}^d_{C,\mathfrak{m}_2} \to \mathrm{Pic}^d_{C,\mathfrak{m}_1}$$

obtained by restricting the isomorphism ψ. Since $\mathfrak{m}_2$ is a finite S-scheme, this map is a surjection as a morphism of étale sheaves. In particular, for any modulus $\mathfrak{m}$, there is a natural surjective morphism of étale sheaves:

$$\mathrm{Pic}^d_{C,\mathfrak{m}} \to \mathrm{Pic}^d_C.$$

Local Freeness and Base Change

Let $\mathfrak{m}$ be a modulus which is everywhere strictly positive. Let g denote the genus of C, which is a locally constant function on S. We restrict our attention to degrees d satisfying the condition:

$$d \geq \max\{2g - 1 + \deg \mathfrak{m}, \deg \mathfrak{m}\}. \tag{3.1}$$

Assuming S is quasi-compact, such a d always exists.

Fix an integer d satisfying the condition above. Let T be an S-scheme and let $\mathcal{L}$ be an invertible sheaf of degree d on C_T. One can show that the pushforwards $\mathrm{pr}_*\mathcal{L}$ and $\mathrm{pr}_*\mathcal{L}(-\mathfrak{m})$ are locally free sheaves and their formations commute with any base change. Explicitly, for any morphism of S-schemes $f : T' \to T$, the base change morphisms are isomorphisms:

$$f^*\mathrm{pr}_*\mathcal{L} \xrightarrow{\sim} \mathrm{pr}_* f^*\mathcal{L}$$

and

$$f^*\mathrm{pr}_*(\mathcal{L}(-\mathfrak{m})) \xrightarrow{\sim} \mathrm{pr}_* f^*(\mathcal{L}(-\mathfrak{m})).$$

In particular, following [Gui19], if $\mathcal{L}$ is an invertible $\mathcal{O}_C$-module with degree d satisfying (3.1) on each fiber of f then $\mathrm{pr}_*\mathcal{L}$ is a locally free $\mathcal{O}_S$-module of rank $d - g + 1$.

Indeed, on each geometric fiber both $\mathcal{L}$ and $\mathcal{L}(-\mathfrak{m})$ have degree at least $2g-1 > \deg(\Omega^1_C) = 2g-2$, so H^1 vanishes on every fiber by Serre duality. Cohomology and base change (Grauert's theorem; see [Mil08]) then gives that $\mathrm{pr}_*\mathcal{L}$ and $\mathrm{pr}_*\mathcal{L}(-\mathfrak{m})$ are locally free and that their formation commutes with arbitrary base change, and Riemann–Roch computes the rank: $h^0(\mathcal{L}) = d-g+1$ and $h^0(\mathcal{L}(-\mathfrak{m})) = d - \deg\mathfrak{m} - g + 1$ on each fiber.

For further background and verification of these constructions, we refer the reader to Milne's notes on abelian varieties ([Mil08]).

3.3 The Abel-Jacobi Morphism and its Fibers

Let $U = C \setminus \mathfrak{m}$ be the complement of the modulus in C. The effective Cartier divisors of degree d which are prime to $\mathfrak{m}$ are parameterized by the symmetric power $\mathrm{Sym}^d_S(U) = U^{(d)}$ over S (see [Gui19] Proposition 4.12, [Mil08] Theorem 3.13). For any such divisor $D \in U^{(d)}$, the associated line bundle $\mathcal{O}_C(D)$ admits a canonical trivialization along $\mathfrak{m}$. Specifically, the canonical section 1_D is regular and non-vanishing on $\mathfrak{m}$ because $\mathrm{supp}(D) \cap \mathrm{supp}(\mathfrak{m}) = \emptyset$. This section restricts to a nowhere-vanishing section on the subscheme $\mathfrak{m}$, thereby determining a trivialization $\psi_D^{-1} : \mathcal{O}_C(D)|_{\mathfrak{m}} \xrightarrow{\sim} \mathcal{O}_{\mathfrak{m}}$. This is done functorially in families, yielding a morphism from the symmetric power to the generalized Picard scheme (over S):

$$\Phi_d : U^{(d)} \to \mathrm{Pic}^d_{C,\mathfrak{m}}, \quad D \mapsto [(\mathcal{O}_C(D), \psi_D)]. \tag{3.2}$$

When $\mathfrak{m} = 0$, $d \geq \max\{2g-1, 0\}$ and C admits a section over S, the symmetric power $C^{(d)}$ is a projective space bundle over Pic^d_C; in particular Φ_d is proper and surjective with geometrically connected fibers.

Guignard ([Gui19] Theorem 4.14) proves that for $\mathfrak{m} > 0$ and d satisfying (3.1), the Abel-Jacobi morphism Φ_d is surjective smooth of relative dimension $d - \deg\mathfrak{m} - g + 1$, with geometrically connected fibers.

When $S = \mathrm{Spec}(k)$, the geometric fibers of Φ_d are well understood:

Theorem 5. *Assume $S = \mathrm{Spec}(k)$ and $d \geq \max\{2g-1+\deg\mathfrak{m}, \deg\mathfrak{m}\}$. Then the geometric fibers of the Abel-Jacobi morphism*

$$\Phi_d : U^{(d)} \to \mathrm{Pic}^d_{C,\mathfrak{m}}$$

over any point are isomorphic to

$$\begin{cases} \mathbb{A}^{d-\deg\mathfrak{m}-g+1}_{k^{sep}} & \textit{if } \mathfrak{m} > 0 \\ \mathbb{P}^{d-g}_{k^{sep}} & \textit{if } \mathfrak{m} = 0 \end{cases}$$

In both cases Φ_d is a fibration in affine spaces or projective spaces, depending on whether $\mathfrak{m}$ is non-zero or zero.

Proof. see [Ten15] Propositions 3.13-3.14, or [Tót11] Prop 2.1.4. □

3.4 Takeuchi's Compactification

We recall that our objective is to descend the local system $\mathcal{F}^{[d]}$ from $U^{(d)}$ to $\mathrm{Pic}^d_{C,\mathfrak{m}}$ along the Abel-Jacobi map Φ_d:

$$\begin{tikzcd} \mathcal{F}^{[d]} \arrow[d] \\ U^{(d)} \arrow[r, "\Phi_d"] & \mathrm{Pic}^d_{C,\mathfrak{m}} \end{tikzcd}$$

(Here, the purple arrow emphasizes that the morphism is of sheaves on the étale site).

However, we encounter an obstruction: in the case we are considering ($\mathfrak{m} > 0$), the fibers of Φ_d are affine spaces (of the same dimension) rather than the better-behaved projective spaces. This hints that a solution to this problem is to compactify the morphism to yield projective fibers.

This section describes the result of the compactification constructed by [Tak19] via the method of blowup.

Let $\mathfrak{m} = \sum_{i=1}^{n} k_P P$ with $\deg P = d_P$ be a modulus on C, and let d satisfy (3.1). Takeuchi ([Tak19]) defines $Z_0 = Z_0(\mathfrak{m}, d)$ as the closed subscheme of $C^{(d)}$ defined by the map $C^{(d-\deg \mathfrak{m})} \to C^{(d)}$ adding $\mathfrak{m}$. He also defines $X_{\mathfrak{m},d}$ as the blowup of $C^{(d)}$ along Z_0. Let $E_0 = E_{\mathfrak{m},d} = Z_0(\mathfrak{m}, d) \times_{C^{(d)}} X_{\mathfrak{m},d}$ be the exceptional divisor of the blowup. It is irreducible of codimension 1, and we let $\eta_0 = \eta_{\mathfrak{m},d}$ be its generic point.

Diagrammatically:

$$\begin{tikzcd} \overline{\{\eta_0\}} = E_0 \arrow[r] \arrow[d] & X_{\mathfrak{m},d} \arrow[d, "\pi"] \\ Z_0 \arrow[r, "c.i"] & C^{(d)} \end{tikzcd}$$

Incorporating $U^{(d)}$, the local system $\mathcal{F}^{[d]}$ and the Abel-Jacobi map, we have:

$$\begin{tikzcd} & & \mathcal{F}^{[d]} \arrow[d] & \\ \overline{\{\eta_0\}} = E_0 \arrow[r] \arrow[d] & X_{\mathfrak{m},d} \arrow[d, "\pi"] & U^{(d)} \arrow[l, hook'] \arrow[r, "\Phi_d"] \arrow[dl, hook'] & \mathrm{Pic}^d_{C,\mathfrak{m}} \\ Z_0 \arrow[r, "c.i"] & C^{(d)} & & \end{tikzcd}$$

In Section 3 of [Tak19] Takeuchi constructs, for d satisfying (3.1), a compactification denoted by $\tilde{C}^{(d)}_{\mathfrak{m}}$ and proves the following:

Theorem 6 (Takeuchi)**.** *The scheme $\tilde{C}^{(d)}_{\mathfrak{m}}$ is an open subscheme of $X_{\mathfrak{m},d}$ containing $U^{(d)}$. The morphism $\Phi_d : U^{(d)} \to \mathrm{Pic}^d_{C,\mathfrak{m}}$ extends to a morphism $\tilde{\Phi}_d : \tilde{C}^{(d)}_{\mathfrak{m}} \to \mathrm{Pic}^d_{C,\mathfrak{m}}$ which makes $\tilde{C}^{(d)}_{\mathfrak{m}}$ a projective space bundle over $\mathrm{Pic}^d_{C,\mathfrak{m}}$. Furthermore, the complement of $U^{(d)}$ in $\tilde{C}^{(d)}_{\mathfrak{m}}$ is isomorphic to the fiber product $E_0 \times_{C^{(d)}} \tilde{C}^{(d)}_{\mathfrak{m}}$.*

Remark (Outline of the construction)*.* We sketch the geometry behind the theorem; for the precise construction see [Tak19, Section 3]. Work over a geometric point of $\mathrm{Pic}^d_{C,\mathfrak{m}}$, i.e. fix a pair $(\mathcal{L}, \psi)$.

The fiber of Φ_d and its natural compactification. A point of the fiber of Φ_d over $(\mathcal{L}, \psi)$ is a divisor $D \in U^{(d)}$ with $(\mathcal{O}_C(D), \psi_D) \cong (\mathcal{L}, \psi)$. Writing $D = \mathrm{div}(s)$ for a section $s \in H^0(C, \mathcal{L})$, the isomorphism condition pins s down to the affine subspace

$$V_\psi \;=\; \{\, s \in H^0(C,\mathcal{L}) \;:\; s|_{\mathfrak{m}} = \psi(1) \,\},$$

a torsor under $W := H^0(C,\mathcal{L}(-\mathfrak{m}))$, which by (3.1) has dimension $N := d - \deg\mathfrak{m} - g + 1$ (see the blue paragraph in Section 3.2). This recovers the affine fibers $\mathbb{A}^N$ of Theorem 5. Inside the full linear system $|\mathcal{L}| = \mathbb{P}(H^0(C,\mathcal{L}))$, the closure of V_ψ is the projective space

$$\mathbb{P}(\overline{V}_\psi), \qquad \overline{V}_\psi := \{\, s \in H^0(C,\mathcal{L}) \;:\; s|_{\mathfrak{m}} \in k\cdot\psi(1) \,\} = W \oplus k\, s_0 \quad (s_0 \in V_\psi \text{ any point}),$$

of the same dimension N. Its boundary $\mathbb{P}(W)$ parametrizes the divisors $D = \mathrm{div}(s)$ with $s|_{\mathfrak{m}} = 0$, i.e. the divisors containing $\mathfrak{m}$ — precisely the points of Z_0 lying in the closure of the fiber.

Why the blowup. The compactified fibers $\mathbb{P}(\overline{V}_\psi)$ for varying ψ (and fixed $\mathcal{L}$) all contain the *same* boundary $\mathbb{P}(W)$: inside $C^{(d)}$ the closures of distinct fibers meet along Z_0, so no open subscheme of $C^{(d)}$ can separate them. Blowing up Z_0 separates these closures: a point of the exceptional divisor E_0 over $D' + \mathfrak{m} \in Z_0$ is a normal direction to Z_0, and for a one-parameter family $D_t \in U^{(d)}$ approaching $D' + \mathfrak{m}$, the direction of approach records the leading term along $\mathfrak{m}$ of the equations of D_t — which is exactly the datum $\lim_t \psi_{D_t}$, up to the scalar already accounted for by projectivizing. Takeuchi defines $\tilde{C}^{(d)}_{\mathfrak{m}}$ as the open subscheme of $X_{\mathfrak{m},d}$ where this leading-term datum is invertible along $\mathfrak{m}$; in particular the strict transforms of the intermediate boundary loci $\{D : \operatorname{supp} D \cap \operatorname{supp}\mathfrak{m} \neq \emptyset,\ D \not\geq \mathfrak{m}\}$ (where no trivialization datum survives) are discarded, which is why the complement of $U^{(d)}$ in $\tilde{C}^{(d)}_{\mathfrak{m}}$ is supported on E_0. On $\tilde{C}^{(d)}_{\mathfrak{m}}$ the pair $(\mathcal{L},\psi)$ can be read off each point, giving the extension $\tilde{\Phi}_d$, and the fiber of $\tilde{\Phi}_d$ over $(\mathcal{L},\psi)$ is the projective space $\mathbb{P}(\overline{V}_\psi) \cong \mathbb{P}^N$ described above, with boundary $\mathbb{P}(W) =$ the fiber of $E_0 \times_{C^{(d)}} \tilde{C}^{(d)}_{\mathfrak{m}}$.

Diagrammatically we have:

$$\begin{tikzcd}
 & & \mathcal{F}^{[d]} \arrow[d, rightsquigarrow] \\
E_0 \times_{C^{(d)}} \tilde{C}^{(d)}_{\mathfrak{m}} \arrow[r] \arrow[d] & \tilde{C}^{(d)}_{\mathfrak{m}} \arrow[d, hook] \arrow[dr, "\tilde{\Phi}_d"] & U^{(d)} \arrow[l, hook'] \arrow[d, "\Phi_d"] \\
\overline{\{\eta_0\}} = E_0 \arrow[r] \arrow[d] & X_{\mathfrak{m},d} \arrow[d, "\pi"] & \mathrm{Pic}^d_{C,\mathfrak{m}} \\
Z_0 \arrow[r, "c.i"] & C^{(d)} &
\end{tikzcd}$$

Also note that $E_0 \times_{C^{(d)}} \tilde{C}^{(d)}_{\mathfrak{m}} = Z_0 \times_{C^{(d)}} \tilde{C}^{(d)}_{\mathfrak{m}}$.

3.5 Tameness Along the Whole Boundary

In this section we prove the technical main theorem of the chapter, which we restate:

Theorem 2. *Let Λ be a finite ring of cardinality invertible in k, and let $\mathcal{F}$ be an étale sheaf of Λ-modules, locally free of rank 1 on U, with ramification bounded by $\mathfrak{m}$. Considering $U^{(d)}$ as an open subscheme of the blowup $\tilde{C}^{(d)}_{\mathfrak{m}}$ of $C^{(d)}$, we have that for sufficiently large integer d, $\mathcal{F}^{[d]}$ is tamely ramified on $H = \tilde{C}^{(d)}_{\mathfrak{m}} \setminus U^{(d)} = E_0 \times_{C^{(d)}} \tilde{C}^{(d)}_{\mathfrak{m}}$.*

The proof reduces Theorem 2 to the single-point calculation of Chapter 2 (Theorem 33), in three steps, working throughout with the G-torsor $\mathcal{P}$ corresponding to $\mathcal{F}$ under Proposition 9:

- Section 3.5.1 proves an auxiliary local result: ramification bounds are stable under external products of torsors on products of blowups.
- Section 3.5.2 shows that the addition maps between symmetric powers of C are étale at the points relevant to us.
- Section 3.5.3 combines the two: it spreads tameness from the exceptional divisor $E_{\mathfrak{n}}$ of a single-point blowup first to sums of coprime submoduli, and then — along the étale addition map — to the generic point η_0 of the full exceptional divisor E_0.

Section 3.5.4 then assembles the proof.

3.5.1 Ramification on Products of Blowups

We begin with an auxiliary local analysis of how ramification bounds behave under products of blowups; it will be used in Section 3.5.3 to combine the contributions of distinct points of the modulus.

Let X and Y be smooth schemes over a field k, and let $x \in X$ and $y \in Y$ be closed points. We denote the blowups of these schemes at the given points by $\pi_X : \mathrm{Bl}_x(X) \to X$ and $\pi_Y : \mathrm{Bl}_y(Y) \to Y$. Furthermore, let $\pi_{X\times Y} : \mathrm{Bl}_{(x,y)}(X \times_k Y) \to X \times_k Y$ be the blowup of the product scheme at the point (x, y). We denote by E_X, E_Y, and $E_{X\times Y}$ the respective exceptional divisors, and let η_X, η_Y, and $\eta_{X\times Y}$ be their generic points.

In this subsection, we establish the following result concerning the stability of ramification bounds under the external product of torsors.

Proposition 7. *Let G be a finite abelian group. Suppose $\mathcal{G}_X$ and $\mathcal{G}_Y$ are G-torsors defined on open subsets $U_X \subset \mathrm{Bl}_x(X)$ and $U_Y \subset \mathrm{Bl}_y(Y)$ that are disjoint from the exceptional divisors. If the ramification of $\mathcal{G}_X$ at η_X and $\mathcal{G}_Y$ at η_Y is bounded by r, then the external product torsor*

$$\mathcal{G}_{X\times Y} := pr_1^{-1}\mathcal{G}_X \otimes pr_2^{-1}\mathcal{G}_Y$$

has ramification bounded by r at the generic point $\eta_{X\times Y}$ of the exceptional divisor in the product blowup.

The proposition is purely local in nature, so it suffices to consider the case where X and Y are affine. More precisely, by the smoothness of X and Y, we may restrict our attention to open neighborhoods of x and y that are étale over affine spaces. The rest of this subsection treats that case.

The Affine Case

Let $X = \mathbb{A}^n_k$ be the affine n-space over a field k, and let $0 \in X$ be the origin. Let $\tilde{X} = \mathrm{Bl}_0(X)$ be the blowup of X at the origin. Recall that $\tilde{X} \subset X \times_k \mathbb{P}^{n-1}_k$ is defined by the equations $x_i u_j = x_j u_i$, where $[u_1 : \cdots : u_n]$ are the homogeneous coordinates of $\mathbb{P}^{n-1}_k$. The exceptional divisor $E \subset \tilde{X}$ is the fiber over the origin, $E = \{(0, [u_1 : \cdots : u_n])\}$, which is of codimension 1 in $\tilde{X}$. Let $\eta \in E$ be the generic point of E, and let $R = \mathcal{O}_{\tilde{X},\eta}$ be the associated local ring. This ring R is a discrete valuation ring (DVR) with fraction field $K = K(X) = k(x_1, \ldots, x_n)$.

On the affine chart U_1 where $u_1 \neq 0$, we have $x_i = \frac{u_i}{u_1} x_1$. The coordinate ring is:

$$\mathcal{O}_{\tilde{X}}(U_1) = k\left[x_1, \frac{u_2}{u_1}, \ldots, \frac{u_n}{u_1}\right]$$

In this chart, the generic point η corresponds to the prime ideal $\mathfrak{p}_1 = (x_1)$. Thus, the local ring is $R = k[x_1, \frac{u_2}{u_1}, \ldots, \frac{u_n}{u_1}]_{(x_1)}$. The residue field is $\kappa(\eta) = k(\frac{u_2}{u_1}, \ldots, \frac{u_n}{u_1})$, and the completion of R with respect to its maximal ideal is:

$$\hat{R} = \kappa(\eta)[[x_1]]$$

In this local ring, x_1 is a uniformizer. Note that any x_i (for $i > 1$) can also serve as a uniformizer, as $x_i = (\frac{u_i}{u_1})x_1$ and $\frac{u_i}{u_1}$ is a unit in R.

For any monomial $M = x_1^{a_1} \ldots x_n^{a_n} \in K$, we can write:

$$M = x_1^{\sum a_i} \left(\frac{u_2}{u_1}\right)^{a_2} \cdots \left(\frac{u_n}{u_1}\right)^{a_n}$$

Since the term in parentheses is a unit in R, the valuation ν_E associated with E satisfies:

$$\nu_E(M) = \sum a_i = \deg M$$

Consequently, for any polynomial $f = f_d + f_{d+1} + \cdots + f_l$, where f_i is the homogeneous part of degree i, we have $\nu_E(f) = d$ (the order of vanishing at the origin).

The Product Case

Now, let $X = \mathbb{A}^n$ and $Y = \mathbb{A}^m$ with origins $x = 0$ and $y = 0$. As before, $\mathrm{Bl}_0(X) \subset X \times \mathbb{P}^{n-1}$ and $\mathrm{Bl}_0(Y) \subset Y \times \mathbb{P}^{m-1}$ have exceptional divisors E_X and E_Y respectively. Consider the product $X \times_k Y \cong \mathbb{A}_k^{n+m}$. The blowup of the product at the origin $(0,0)$, denoted $\mathrm{Bl}_{(0,0)}(X \times_k Y)$, is a subscheme of $(X \times Y) \times \mathbb{P}^{n+m-1}$ defined by:

$$\begin{cases} x_i w_j = x_j w_i & 1 \leq i, j \leq n \\ y_k w_{n+l} = y_l w_{n+k} & 1 \leq k, l \leq m \\ x_i w_{n+j} = y_j w_i & 1 \leq i \leq n, \ 1 \leq j \leq m \end{cases}$$

where $[w_1 : \cdots : w_{n+m}]$ are the homogeneous coordinates of $\mathbb{P}^{n+m-1}$. The exceptional divisor $E_{X \times Y}$ is isomorphic to $\mathbb{P}^{n+m-1}$.

Comparison of Blowups

Both $\mathrm{Bl}_0(X) \times_k \mathrm{Bl}_0(Y)$ and $\mathrm{Bl}_{(0,0)}(X \times_k Y)$ are birational to $X \times_k Y$. They share a common dense open set $\tilde{U}$ defined by the condition that neither the X-coordinates nor the Y-coordinates vanish simultaneously in the projective space:

$$\tilde{U} = \{((x,y), [w_1 : \cdots : w_{n+m}]) \mid (w_1, \ldots, w_n) \neq 0 \text{ and } (w_{n+1}, \ldots, w_{n+m}) \neq 0\}$$

This yields a diagram of open immersions:

$$\begin{tikzcd}
& \tilde{U} \arrow[dl, hook', "f_1"'] \arrow[dr, hook, "f_2"] & \\
\mathrm{Bl}_0(X) \times_k \mathrm{Bl}_0(Y) & & \mathrm{Bl}_{(0,0)}(X \times_k Y)
\end{tikzcd}$$

where f_1 maps the coordinates to the respective projectivizations $[w_1 : \cdots : w_n]$ and $[w_{n+1} : \cdots : w_{n+m}]$. Also note that the generic point $\eta_{X \times Y}$ of $E_{X \times_k Y}$ is inside $\tilde{U}$.

Extensions of DVRs

Let S be the local ring of the generic point $\eta_{X\times Y}$ of $E_{X\times Y}$ in $\tilde{U}$. On the chart where $w_1 \neq 0$ and $w_{n+1} \neq 0$, we have $x_1 = (\frac{w_1}{w_{n+1}})y_1$. Since $\frac{w_1}{w_{n+1}}$ is a unit in this chart, x_1 and y_1 are equivalent as uniformizers. We have:

$$S = k\left[x_1, \frac{w_2}{w_1} \cdots, \frac{w_n}{w_1}, \frac{w_{n+1}}{w_1} \cdots \frac{w_{n+m}}{w_1}\right]_{(x_1)} = k\left[\frac{w_1}{w_{n+1}}, \frac{w_2}{w_{n+1}} \cdots \frac{w_n}{w_{n+1}}, y_1 \cdots \frac{w_{n+m}}{w_{n+1}}\right]_{(y_1)}$$

$$k(\eta_{X\times Y}) = k\left(\frac{w_2}{w_1} \cdots, \frac{w_n}{w_1}, \frac{w_{n+1}}{w_1} \cdots \frac{w_{n+m}}{w_1}\right)$$

Let R_X be the local ring of the exceptional divisor E_X in $\mathrm{Bl}_0(X)$. The pullback of $E_X \times_k \mathrm{Bl}_0(Y)$ along f_1 induces an extension of DVRs $R_X \hookrightarrow S$. Which is:

1. Weakly Unramified: x_1 is a uniformizer in both R_X and S, so the ramification index is $e = 1$.
2. Residually Transcendental: The residue field extension $\kappa(\eta_X) \subset \kappa(\eta)$ is:
$$k\left(\frac{w_2}{w_1}, \ldots, \frac{w_n}{w_1}\right) \subset k\left(\frac{w_2}{w_1}, \ldots, \frac{w_n}{w_1}, \frac{w_{n+1}}{w_1}, \ldots, \frac{w_{n+m}}{w_1}\right)$$
Hence separable.

Since this extension is generated by transcendental elements, it is separable and formally smooth at the maximal ideal ([Stacks, Tag 09E7]).

Ramification of G-Torsors

Let G be a finite abelian group. Let $\mathcal{Q}$ be a G-torsor on an open $U \subset \mathrm{Bl}_0(X)$ disjoint from E_X. Let $V \subset \mathrm{Bl}_0(Y)$ be an open subscheme, and let $\pi_X : \mathrm{Bl}_0(X) \times_k V \to \mathrm{Bl}_0(X)$ be the projection onto the first factor. By restricting this projection to $U \times_k V$, we obtain the pullback G-torsor:

$$\pi_X^{-1}(\mathcal{Q}) \cong \mathcal{Q} \times_k V$$

which is defined on the open subset $U \times_k V \subset \mathrm{Bl}_0(X) \times_k \mathrm{Bl}_0(Y)$. The extension of local rings $R_X \to S$ is weakly unramified (the ramification index $e = 1$) and residually transcendental with separable residue field extension. Under these conditions the ramification filtration is preserved. Therefore, the pullback $\pi_X^{-1}(\mathcal{Q})$ has ramification bounded by r at the generic point $\eta_{X\times Y}$ of the exceptional divisor $E_{X\times Y}$ if and only if the original torsor $\mathcal{Q}$ has ramification bounded by r at the generic point η_X of E_X.

And we finish by Lemma 42.

3.5.2 Étaleness of the Addition Maps

The following lemma is adapted from [Tak19] (Lemma 4.1).

Lemma 8. *Let C be a projective, smooth, and geometrically connected curve over a perfect field k. Let $\mathfrak{m} = \sum_{i=1}^{r} k_i P_i$ be an effective divisor where $P_1, \ldots, P_r$ are distinct closed points. Let $U = C \setminus \mathfrak{m}$ and let $d \geq \deg \mathfrak{m}$.*

Suppose $\mathfrak{n}_1, \ldots, \mathfrak{n}_l$ are pairwise coprime submoduli of $\mathfrak{m}$ such that $\mathfrak{m} = \sum_{j=1}^{l} \mathfrak{n}_j$. Consider the summation morphism:

$$\pi : C^{(\deg \mathfrak{n}_1)} \times_k \cdots \times_k C^{(\deg \mathfrak{n}_l)} \times_k C^{(d-\deg \mathfrak{m})} \longrightarrow C^{(d)}$$

defined by $(D_1, \ldots, D_l, D_{extra}) \mapsto \sum_{j=1}^{l} D_j + D_{extra}$.

Then π is étale at the generic point of the closed subvariety

$$V = \{\mathfrak{n}_1\} \times_k \cdots \times_k \{\mathfrak{n}_l\} \times_k C^{(d-\deg \mathfrak{m})}$$

inside the domain $C^{(\deg \mathfrak{n}_1)} \times_k \cdots \times_k C^{(\deg \mathfrak{n}_l)} \times_k C^{(d-\deg \mathfrak{m})}$.

Proof. We may assume that k is algebraically closed (hence $\deg P_i = 1$ for all i). By miracle flatness π is flat; it is quasi-finite and projective as a map between projective spaces, so we conclude π is finite and flat. It is enough to show that there exists a closed point Q of $\mathfrak{n}_1 + \cdots + \mathfrak{n}_l + C^{(d-\deg \mathfrak{m})} \subset C^{(d)}$ over which there are $\deg \pi$ points on $C^{(\mathfrak{n}_1)} \times_k \cdots \times_k C^{(\mathfrak{n}_l)} \times_k C^{(d-\deg \mathfrak{m})}$. (Because it will be unramified at this point and thus also at the generic point of V.) Choose Q as a point corresponding to a divisor $\mathfrak{n}_1 + \cdots + \mathfrak{n}_l + P_{r+1} + \cdots + P_{r+d-\deg \mathfrak{m}}$, where $P_1, \ldots, P_{r+d-\deg \mathfrak{m}}$ are distinct points of $U(k)$. $\square$

Corollary 9. *The morphism $C^{(\deg \mathfrak{m})} \times_k C^{(d-\deg \mathfrak{m})} \xrightarrow{\pi} C^{(d)}$ is finite flat everywhere, and étale at the generic point of the closed subvariety $Z_\mathfrak{m} \times_k C^{(d-\deg \mathfrak{m})} \subset C^{(\deg \mathfrak{m})} \times_k C^{(d-\deg \mathfrak{m})}$.*

Proof. This is the case $l = 1$, $\mathfrak{n}_1 = \mathfrak{m}$ of Lemma 8 (note that $Z_\mathfrak{m}$ is the single point $\{\mathfrak{m}\}$ of $C^{(\deg \mathfrak{m})}$); finiteness and flatness everywhere were established in the first paragraph of its proof. $\square$

3.5.3 Spreading Tameness from $\eta_\mathfrak{m}$ to η_0

Following from Corollary 9, we look at the following diagram, coming from the flat base change $C^{(d-\deg \mathfrak{m})} \to \mathrm{Spec}(k)$ (Proposition 20) of (2.1):

$$\begin{tikzcd}
E_\mathfrak{m} \times_k C^{(d-\deg \mathfrak{m})} \arrow[r] \arrow[d] & X_\mathfrak{m} \times_k C^{(d-\deg \mathfrak{m})} \arrow[d] & \mathcal{F}^{[\deg \mathfrak{m}]} \times_k C^{(d-\deg \mathfrak{m})} \arrow[d] \\
Z_\mathfrak{m} \times_k C^{(d-\deg \mathfrak{m})} \arrow[r] & C^{(\deg \mathfrak{m})} \times_k C^{(d-\deg \mathfrak{m})} & U^{(\deg \mathfrak{m})} \times_k C^{(d-\deg \mathfrak{m})} \arrow[ul]
\end{tikzcd}$$

Note that $U^{(\deg \mathfrak{m})} \times_k C^{(d-\deg \mathfrak{m})}$ is a dense open subscheme of $X_\mathfrak{m} \times_k C^{(d-\deg \mathfrak{m})}$, and $E_\mathfrak{m} \times_k C^{(d-\deg \mathfrak{m})}$ is a prime divisor of $X_\mathfrak{m} \times_k C^{(d-\deg \mathfrak{m})}$. Hence it is a well defined question according to Definition 38 to ask whether $\mathcal{F}^{[\deg \mathfrak{m}]} \times_k C^{(d-\deg \mathfrak{m})}$ is tamely ramified at the generic point θ of $E_\mathfrak{m} \times_k C^{(d-\deg \mathfrak{m})}$.

Lemma 10. *If $\mathcal{F}^{[\deg \mathfrak{m}]}$ is tamely ramified at $\eta_{\mathfrak{m}}$, then $\mathcal{F}^{[\deg \mathfrak{m}]} \times_k C^{(d-\deg \mathfrak{m})}$ is tamely ramified at the generic point θ of $E_{\mathfrak{m}} \times_k C^{(d-\deg \mathfrak{m})}$.*

Proof. This follows from [Stacks, Tag 0EYD] □

Replacing $C^{(d-\deg \mathfrak{m})}$ with the dense open subscheme $U^{(d-\deg \mathfrak{m})} \subset C^{(d-\deg \mathfrak{m})}$, Lemma 10 gives that the G-torsor $p_1^{-1}\mathcal{P}^{[\deg \mathfrak{m}]}$ ($\mathcal{P}$ corresponds to $\mathcal{F}$ under Proposition 9) is tamely ramified at θ the generic point of $E_{\mathfrak{m}} \times_k U^{(d-\deg \mathfrak{m})} \subset U^{(\deg \mathfrak{m})} \times_k U^{(d-\deg \mathfrak{m})}$, where $p_1 : U^{(\deg \mathfrak{m})} \times_k U^{(d-\deg \mathfrak{m})} \to U^{(\deg \mathfrak{m})}$ is the projection to the first factor.

Looking at the second projection $p_2 : X_{\mathfrak{m}} \times_k U^{(d-\deg \mathfrak{m})} \to U^{(d-\deg \mathfrak{m})}$, and the fact that $\mathcal{P}^{[d-\deg \mathfrak{m}]}$ is étale on $U^{(d-\deg \mathfrak{m})}$ we get that $p_2^{-1}\mathcal{P}^{[d-\deg \mathfrak{m}]} = X_{\mathfrak{m}} \times_k \mathcal{P}^{[d-\deg \mathfrak{m}]}$ is étale on $X_{\mathfrak{m}} \times_k U^{(d-\deg \mathfrak{m})}$. Hence, its restriction to $U^{(\deg \mathfrak{m})} \times_k U^{(d-\deg \mathfrak{m})}$ is unramified at θ.

Thus, by Lemma 11 below, we conclude that $\mathcal{P}^{[\deg \mathfrak{m}]} \boxtimes \mathcal{P}^{[d-\deg \mathfrak{m}]} = p_1^{-1}\mathcal{P}^{[\deg \mathfrak{m}]} \otimes^G p_2^{-1}\mathcal{P}^{[d-\deg \mathfrak{m}]}$ is tamely ramified at θ the generic point of $E_{\mathfrak{m}} \times_k U^{(d-\deg \mathfrak{m})}$.

Lemma 11. *Let $X \to \operatorname{Spec} k$ be a scheme over a field k, and let $\mathcal{P}_1, \mathcal{P}_2$ be two G-torsors on U_{et}. Let ξ be the generic point of a prime divisor $D \subset X$. If $\mathcal{P}_1$ is tamely ramified at ξ, and $\mathcal{P}_2$ is unramified at ξ, then the product $\mathcal{P}_1 \otimes^G \mathcal{P}_2$ is tamely ramified at ξ.*

Proof. This follows from Lemma 42. □

Combining this with Corollary 9 we get

Corollary 12. *Let $\mathfrak{m}$ be a modulus as above, and let $\eta_{\mathfrak{m}}$ be the generic point of $E_{\mathfrak{m}}$. Let $\mathcal{P}$ be a G-torsor on U_{et} with ramification bounded by $\mathfrak{m}$. Assume $\mathcal{P}^{[\deg \mathfrak{m}]}$ is tamely ramified at $\eta_{\mathfrak{m}}$. Then $\mathcal{P}^{[\deg \mathfrak{m}]} \boxtimes \mathcal{P}^{[d-\deg \mathfrak{m}]}$ is tamely ramified at the generic point θ of $E_{\mathfrak{m}} \times_k C^{(d-\deg \mathfrak{m})} \subset C^{(\deg \mathfrak{m})} \times_k C^{(d-\deg \mathfrak{m})}$, and $\mathcal{P}^{[d]}$ is tamely ramified at the generic point $\eta_0 = \eta_{\mathfrak{m},d}$ of $E_0 = E_{\mathfrak{m},d}$*

Proof. The first assertion, that $\mathcal{P}^{[\deg \mathfrak{m}]} \boxtimes \mathcal{P}^{[d-\deg \mathfrak{m}]}$ is tamely ramified at θ, follows from the preceding discussion. Thus, it remains to show that $\mathcal{P}^{[d]}$ is tamely ramified at η_0.

Consider the blowup diagram defining $X_{\mathfrak{m},d}$:

$$\begin{tikzcd}
\overline{\{\eta_0\}} = E_0 \arrow[r] \arrow[d] & X_{\mathfrak{m},d} \arrow[d, "\pi"] \\
Z_0 \arrow[r, "c.i"] & C^{(d)}
\end{tikzcd}$$

By performing a base change along the flat addition map $+ : C^{(\deg \mathfrak{m})} \times_k U^{(d-\deg \mathfrak{m})} \to C^{(d)}$, we obtain the following commutative diagram:

$$\begin{tikzcd}
 & \left(C^{(\deg \mathfrak{m})} \times_k U^{(d-\deg \mathfrak{m})}\right) \times_{C^{(d)}} E_0 \arrow[r] \arrow[d] & \overline{\{\eta_0\}} = E_0 \arrow[d] \\
 & \left(C^{(\deg \mathfrak{m})} \times_k U^{(d-\deg \mathfrak{m})}\right) \times_{C^{(d)}} X_{\mathfrak{m},d} \arrow[r] \arrow[d] & X_{\mathfrak{m},d} \arrow[d, "\pi"] \\
\mathfrak{m} \times_k U^{(d-\deg \mathfrak{m})} \arrow[r, "c.i"] & C^{(\deg \mathfrak{m})} \times_k U^{(d-\deg \mathfrak{m})} \arrow[r, "+"] & C^{(d)}
\end{tikzcd}$$

By Theorem 30, blowups commute with flat base change. Since the inverse image of the center Z_0 under the map $+$ is $\mathfrak{m} \times_k U^{(d-\deg \mathfrak{m})}$, the scheme $\left(C^{(\deg \mathfrak{m})} \times_k U^{(d-\deg \mathfrak{m})}\right) \times_{C^{(d)}} X_{\mathfrak{m},d}$ is the blowup of $C^{(\deg \mathfrak{m})} \times_k U^{(d-\deg \mathfrak{m})}$ along $\mathfrak{m} \times_k U^{(d-\deg \mathfrak{m})}$. This, in turn, is isomorphic to the base change of the blowup $X_{\mathfrak{m}}$ (of $C^{(\deg \mathfrak{m})}$ along $\mathfrak{m}$) via the (flat) projection $C^{(\deg \mathfrak{m})} \times_k U^{(d-\deg \mathfrak{m})} \to C^{(\deg \mathfrak{m})}$.

Assembling these facts, we obtain the following Cartesian square:

$$
\begin{tikzcd}
 & E_{\mathfrak{m}} \times U^{(d-\deg \mathfrak{m})} \arrow[r, "\tilde{+}"] \arrow[d] & \overline{\{\eta_0\}} = E_0 \arrow[d] \\
 & X_{\mathfrak{m}} \times_k U^{(d-\deg \mathfrak{m})} \arrow[r, "\tilde{+}"] \arrow[d] & X_{\mathfrak{m},d} \arrow[d, "\pi"] \\
\mathfrak{m} \times_k U^{(d-\deg \mathfrak{m})} \arrow[r, "c.i"] & C^{(\deg \mathfrak{m})} \times_k U^{(d-\deg \mathfrak{m})} \arrow[r, "+"] & C^{(d)}
\end{tikzcd}
$$

The point θ defined in the Corollary is the generic point of $E_{\mathfrak{m}} \times_k U^{(d-\deg \mathfrak{m})}$. Let η be the generic point of $\mathfrak{m} \times_k U^{(d-\deg \mathfrak{m})}$. By Corollary 9, the map $+$ is étale at η. Consequently, the lifted map $\tilde{+}$ is étale at θ. Given the isomorphism $(+^{-1})(\mathcal{P}^{[d]}) \cong \mathcal{P}^{[\deg \mathfrak{m}]} \boxtimes \mathcal{P}^{[d-\deg \mathfrak{m}]}$ from Proposition 28, the tame ramification of the box product at θ descends to the tame ramification of $\mathcal{P}^{[d]}$ at η_0 by applying Lemma 43.

□

Lemma 13. *Let $\mathfrak{n}_1, \mathfrak{n}_2 \subset \mathfrak{m}$ be two coprime sub moduli of $\mathfrak{m}$. Assume $\mathcal{P}^{[\deg \mathfrak{n}_1]}$, $\mathcal{P}^{[\deg \mathfrak{n}_2]}$ are at most tamely ramified at $\eta_{\mathfrak{n}_1}$, $\eta_{\mathfrak{n}_2}$ respectively. Then $\mathcal{P}^{[\deg \mathfrak{n}_1 + \deg \mathfrak{n}_2]}$ is at most tamely ramified at $\eta_{\mathfrak{n}_1+\mathfrak{n}_2}$.*

Proof. It follows from Proposition 7 and Proposition 28. □

3.5.4 Proof of Theorem 2

Proof of Theorem 2. Let $\mathcal{F}$ be as in Theorem 2, $\mathfrak{m} = \sum_{i=1}^n k_P P$ with $\deg P = d_P$. Then by Theorem 3, for every $\mathfrak{n} \subset \mathfrak{m}$ of the form $\mathfrak{n} = k_P P$, $\mathcal{F}^{[\deg \mathfrak{n}]}$ is at most tamely ramified at $\eta_{\mathfrak{n}}$. By Lemma 13, $\mathcal{F}^{[\deg \mathfrak{m}]}$ is then at most tamely ramified at $\eta_{\mathfrak{m}}$. And thus by Corollary 12, $\mathcal{F}^{[d]}$ is tamely ramified at the generic point η_0 of E_0. □

3.6 Proof of Theorem 1

We work over $S = \operatorname{Spec} k$, for k perfect.

By Proposition 9, it is equivalent to prove the following torsor formulation.

Theorem 14. *Let $G = \Lambda^\times$ be a finite abelian group (Λ as before), and let $\mathcal{P}$ be a G-torsor on U, with ramification bounded by $\mathfrak{m}$. Then, for sufficiently large integer d, there exists a unique (up to isomorphism) G-torsor $\mathcal{Q}_d$ on $\operatorname{Pic}^d_{C,\mathfrak{m}}$, such that the pullback of $\mathcal{Q}_d$ by Φ_d is isomorphic to $\mathcal{P}^{[d]}$.*

Proof. Let $\mathcal{F} := \mathcal{P} \times^G \Lambda$ be the rank-one local system associated with $\mathcal{P}$ under Proposition 9; by Definition 26, the frame torsor of $\mathcal{F}^{[d]}$ is $\mathcal{P}^{[d]}$, so statements about $\mathcal{F}^{[d]}$ and about $\mathcal{P}^{[d]}$ translate into one another. We divide the proof into two cases, when $\mathfrak{m} = 0$ and when $\mathfrak{m} > 0$.

Case 1: $\mathfrak{m} = 0$. By Section 3.3, for d large enough, the Abel-Jacobi morphism $\Phi_d : C^{(d)} \to \mathrm{Pic}^d_C$ is proper surjective and smooth, with geometrically connected fibers, each isomorphic to $\mathbb{P}^{d-g}_{k^{sep}}$ (Theorem 5). Hence by Corollary 4 it induces an exact sequence of etale fundamental groups:

$$\pi_1^{et}(\mathbb{P}^{d-g}_{k^{sep}}) \to \pi_1^{et}(C^{(d)}) \to \pi_1^{et}(\mathrm{Pic}^d_C) \to 1$$

But $\mathbb{P}^{d-g}_{k^{sep}}$ is simply connected ([Ten15] Example 4.9, [Tót11] Example 1.4.12), hence its etale fundamental group is trivial, and we get an isomorphism of etale fundamental groups:

$$\pi_1^{et}(C^{(d)}) \cong \pi_1^{et}(\mathrm{Pic}^d_C)$$

Implying the theorem in this case.

Case 2: $\mathfrak{m} > 0$. In this case by Theorem 6, for d large enough, the Abel-Jacobi morphism $\Phi_d : U^{(d)} \to \mathrm{Pic}^d_{C,\mathfrak{m}}$ extends to a proper surjective and smooth map, with geometrically connected fibers isomorphic to projective spaces,

$$\tilde{\Phi}_d : \tilde{C}^{(d)}_{\mathfrak{m}} \to \mathrm{Pic}^d_{C,\mathfrak{m}}$$

Hence we get an isomorphism of etale fundamental groups:

$$\pi_1^{et}(\tilde{C}^{(d)}_{\mathfrak{m}}) \cong \pi_1^{et}(\mathrm{Pic}^d_{C,\mathfrak{m}})$$

By Theorem 2, $\mathcal{F}^{[d]}$ is tamely ramified on the boundary divisor $H = \tilde{C}^{(d)}_{\mathfrak{m}} \setminus U^{(d)}$.

Thus, by Lemma 15 below, we have: $\mathcal{F}^{[d]}$ extends to a locally constant sheaf $\tilde{\mathcal{F}}^{[d]}$ on $\tilde{C}^{(d)}_{\mathfrak{m}}$, which by the isomorphism of etale fundamental groups above, corresponds to a unique locally constant sheaf $\mathcal{G}_d$ on $\mathrm{Pic}^d_{C,\mathfrak{m}}$, such that $\tilde{\Phi}_d^* \mathcal{G}_d \cong \tilde{\mathcal{F}}^{[d]}$. Restricting back to $U^{(d)}$, we get $\Phi_d^* \mathcal{G}_d \cong \mathcal{F}^{[d]}$, as required in the local-system formulation. Finally, let $\mathcal{Q}_d = \mathrm{Fr}(\mathcal{G}_d)$. Since the frame construction commutes with pullback, the preceding isomorphism and Definition 26 give $\Phi_d^* \mathcal{Q}_d \cong \mathcal{P}^{[d]}$. Uniqueness follows from the equivalence in Proposition 9. □

Lemma 15. *If $\mathcal{F}^{[d]}$ is a locally constant sheaf on $U^{(d)}$ which is tamely ramified along the boundary divisor $H = \tilde{C}^{(d)}_{\mathfrak{m}} \setminus U^{(d)}$, then $\mathcal{F}^{[d]}$ extends to a locally constant sheaf $\tilde{\mathcal{F}}^{[d]}$ on $\tilde{C}^{(d)}_{\mathfrak{m}}$.*

Proof. The lemma we are referencing above can be proved in two routes:
Route 1 - Showing $\ker\{\pi_1^{ab}(U^{(d)}) \to \pi_1^{ab}(\mathrm{Pic}^d_{C,\mathfrak{m}})\}$ is pro-p group.
Route 2 - Showing

$$\pi_1^{t,ab}(U^{(d)}) \to \pi_1^{t,ab}(\mathrm{Pic}^d_{C,\mathfrak{m}})$$

is isomorphism to its image. here one needs to be precise.

□

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